Lab Report: IPv4 & IPv6 Troubleshoot Static and Default Routes

Activity 16.3.1

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This report documents the identification and remediation of routing misconfigurations across a multi-router Cisco Packet Tracer topology. The lab involved verifying IPv4 and IPv6 connectivity between end hosts (PC1, PC2) and a central server, identifying incorrect default/static routes, and applying corrective commands on routers R1, R2, and R3.

1 Network Topology Overview

The initial phase involved checking the IPv4 and IPv6 address configurations on PC1, PC2, and the Server to establish baseline parameters.

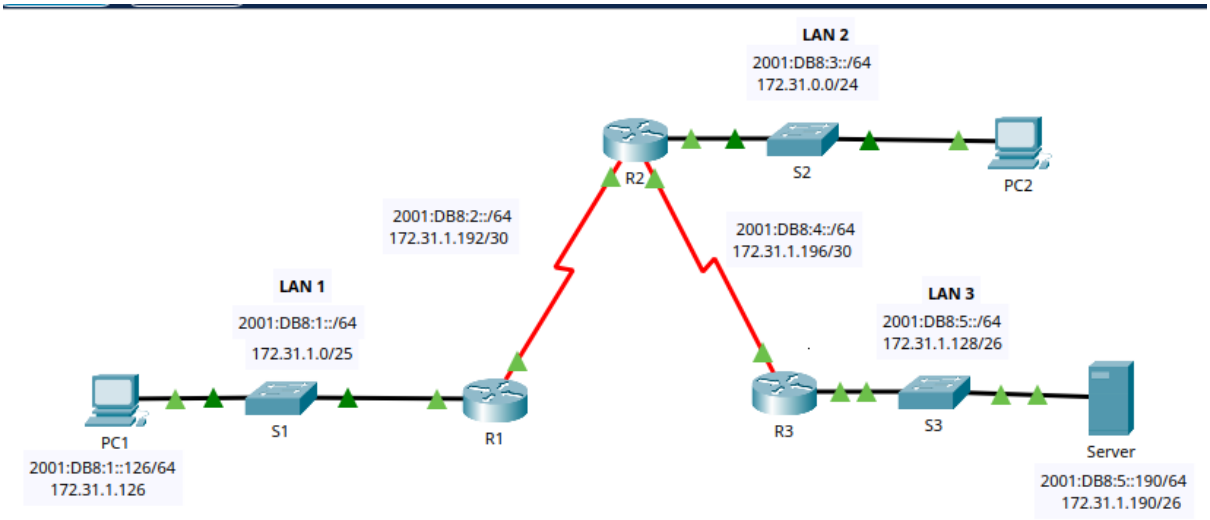


Figure 1: Overall Network Topology in Cisco Packet Tracer.

The network consists of three routers ($R1, R2, R3$) connected via point-to-point serial links, with local area networks attached to each router.

2 Router R1 Configuration & Troubleshooting

2.1 Issue Identification

Initial connectivity testing began by pinging $R2$ from $PC1$, which failed. A review of the routing tables on $R1$ revealed two primary routing errors:

```
Gateway of last resort is 172.31.1.195 to network 0.0.0.0

     172.31.0.0/16 is variably subnetted, 4 subnets, 3 masks
C       172.31.1.0/25 is directly connected, GigabitEthernet0/0
L       172.31.1.1/32 is directly connected, GigabitEthernet0/0
C       172.31.1.192/30 is directly connected, Serial0/0/0
L       172.31.1.194/32 is directly connected, Serial0/0/0
S*    0.0.0.0/0 [1/0] via 172.31.1.195
```

Figure 2: R1 IPv4 Routing Table Output.

1. **IPv4 Next-Hop Misconfiguration:** As shown in Figure 2, the current default route (`0.0.0.0/0`) pointed to `172.31.1.195`. Since $R1$'s Serial0/0/0 interface is `172.31.1.194/30`, the correct next-hop address on $R2$'s side is `172.31.1.193`.
2. **IPv6 Missing Default Route:** As shown in Figure 3, $R1$ contained three explicit static routes (`2001:DB8:3::/64`, `4::/64`, and `5::/64`) instead of a default route. Because $R1$ is a stub router with only one path to the rest of the network via $R2$, individual static routes add unnecessary complexity.

```
C   2001:DB8:1::/64 [0/0]
     via GigabitEthernet0/0, directly connected
L   2001:DB8:1::1/128 [0/0]
     via GigabitEthernet0/0, receive
C   2001:DB8:2::/64 [0/0]
     via Serial0/0/0, directly connected
L   2001:DB8:2::194/128 [0/0]
     via Serial0/0/0, receive
S   2001:DB8:3::/64 [1/0]
     via Serial0/0/0, directly connected
S   2001:DB8:4::/64 [1/0]
     via Serial0/0/0, directly connected
S   2001:DB8:5::/64 [1/0]
     via Serial0/0/0, directly connected
L   FF00::/8 [0/0]
     via Null0, receive
R1#
```

Figure 3: R1 IPv6 Routing Table Output.

2.2 Remediation

The incorrect IPv4 default route and unnecessary IPv6 static routes were removed, followed by the addition of proper IPv4 and IPv6 default routes targeting *R*2:

```
! Correct IPv4 Default Route
R1(config)# no ip route 0.0.0.0 0.0.0.0 172.31.1.195
R1(config)# ip route 0.0.0.0 0.0.0.0 172.31.1.193

! Remove Inefficient IPv6 Static Routes
R1(config)# no ipv6 route 2001:DB8:3::/64 Serial0/0/0
R1(config)# no ipv6 route 2001:DB8:4::/64 Serial0/0/0
R1(config)# no ipv6 route 2001:DB8:5::/64 Serial0/0/0

! Add IPv6 Default Route
R1(config)# ipv6 route ::/0 2001:DB8:2::193
```

Listing 1: R1 Corrective Configuration

3 Router R2 Configuration & Troubleshooting

3.1 Issue Identification

*R*2 acts as the central router connecting *R*1 (Serial0/0/0, `172.31.1.194`) and *R*3 (Serial0/0/1, `172.31.1.198`).

```
Device Name: R2
Device Model: 1941
Hostname: R2

Port                Link   VLAN   IP Address         IPv6 Address                              MAC Address
GigabitEthernet0/0  Up     --     172.31.0.1/24      2001:DB8:3::1/64                          00E0.8F4D.0A01
GigabitEthernet0/1  Down   --     <not set>          <not set>                                 00E0.8F4D.0A02
Serial0/0/0         Up     --     172.31.1.193/30    2001:DB8:2::193/64                        <not set>
Serial0/0/1         Up     --     172.31.1.197/30    2001:DB8:4::197/64                        <not set>
Vlan1               Down   1      <not set>          <not set>                                 0090.2112.DA7B
```

Figure 4: R2 Interface Overview in Packet Tracer.

Inspection of $R2$'s routing tables revealed three critical routing errors:

1. **Inverted IPv4 Next-Hops:**
   - The route for $R1$'s LAN (`172.31.1.0/25`) was incorrectly pointing to $R3$ (`172.31.1.198`).
   - The route for $R3$'s LAN (`172.31.1.128/26`) was incorrectly pointing to $R1$ (`172.31.1.194`).

```
Gateway of last resort is not set

     172.31.0.0/16 is variably subnetted, 8 subnets, 5 masks
C       172.31.0.0/24 is directly connected, GigabitEthernet0/0
L       172.31.0.1/32 is directly connected, GigabitEthernet0/0
S       172.31.1.0/25 [1/0] via 172.31.1.198
S       172.31.1.128/26 [1/0] via 172.31.1.194
C       172.31.1.192/30 is directly connected, Serial0/0/0
L       172.31.1.193/32 is directly connected, Serial0/0/0
C       172.31.1.196/30 is directly connected, Serial0/0/1
L       172.31.1.197/32 is directly connected, Serial0/0/1
```

Figure 5: R2 IPv4 Static Route Analysis.

2. **IPv6 Subnet Typo:** The route intended for $R1$'s IPv6 LAN was configured as `2001:DB6:1::/64` instead of `2001:DB8:1::/64` (Figure 6).

```
S   2001:DB6:1::/64 [1/0]
     via 2001:DB8:2::194
C   2001:DB8:2::/64 [0/0]
     via Serial0/0/0, directly connected
L   2001:DB8:2::193/128 [0/0]
     via Serial0/0/0, receive
C   2001:DB8:3::/64 [0/0]
     via GigabitEthernet0/0, directly connected
L   2001:DB8:3::1/128 [0/0]
     via GigabitEthernet0/0, receive
C   2001:DB8:4::/64 [0/0]
     via Serial0/0/1, directly connected
L   2001:DB8:4::197/128 [0/0]
     via Serial0/0/1, receive
S   2001:DB8:5::/64 [1/0]
     via 2001:DB8:4::198
L   FF00::/8 [0/0]
     via Null0, receive
R2#
```

Figure 6: R2 IPv6 Misconfiguration Output.

3.2 Remediation

The routes with swapped next-hops and the mistyped IPv6 prefix were removed and reconfigured with correct parameters:

```
R2# configure terminal
! Fix Swapped IPv4 Routes
R2(config)# no ip route 172.31.1.0 255.255.255.128 172.31.1.198
R2(config)# no ip route 172.31.1.128 255.255.255.192 172.31.1.194
R2(config)# ip route 172.31.1.0 255.255.255.128 172.31.1.194
R2(config)# ip route 172.31.1.128 255.255.255.192 172.31.1.198

! Fix IPv6 Typo
R2(config)# no ipv6 route 2001:DB6:1::/64 2001:DB8:2::194
R2(config)# ipv6 route 2001:DB8:1::/64 2001:DB8:2::194
R2(config)# end
```

Listing 2: R2 Corrective Configuration

3.3 Midway Verification

Following changes on $R1$ and $R2$, immediate connectivity tests verified successful communication across the left portion of the network.

Fire	Last Status	Source	Destination	Type	Color	Time(sec)
	Successful	PC1	PC2	ICMP		0.000
	Successful	PC1	R2	ICMP		0.000
	Successful	R1	PC2	ICMP		0.000

Figure 7: Intermediate Connectivity Check (PC1 to R2, PC1 to PC2, PC2 to R1).

4 Router R3 Configuration & Troubleshooting

4.1 Issue Identification

Testing connectivity toward the Server revealed that while $PC2 \rightarrow Server$, $Server \rightarrow R2$, and $Server \rightarrow R1$ were successful, traffic from $Server \rightarrow PC1$ failed.

Fire	Last Status	Source	Destination	Type	Color	Time(sec)
	Successful	PC2	Server	ICMP		0.000
	Successful	R2	Server	ICMP		0.000
	Successful	Server	R1	ICMP		0.000
	Failed	Server	PC1	ICMP		0.000

Figure 8: Initial Server Ping Diagnostics.

Analysis of $R3$'s configurations identified the following issues:

```
R3#show ip route static
     172.31.0.0/16 is variably subnetted, 7 subnets, 5 masks
S       172.31.0.0/24 is directly connected, Serial0/0/1
S       172.31.1.0/28 is directly connected, Serial0/0/1
S       172.31.1.192/30 is directly connected, Serial0/0/1
```

Figure 9: R3 IPv4 Static Routes.

1. **IPv4 Subnet Mask & Structure Errors:**
   - $R3$ used multiple individual static routes pointing out Serial0/0/1 despite being a stub router that only requires a single default route.
   - The static route for $R1$'s LAN (`172.31.1.0`) used an incorrect `/28` (`255.255.255.240`) mask instead of the required `/25` (`255.255.255.128`) mask.
2. **IPv6 Omissions & Redundancies:**
   - Missing route for $R1$'s LAN (`2001:DB8:1::/64`).
   - Unnecessary route pointing to $R2$'s LAN (`2001:DB8:3::/64`) over the serial link.
   - Lack of a default IPv6 route (`::/0`).

```
      D - EIGRP, EX - EIGRP external
S   2001:DB8:2::/64 [1/0]
     via Serial0/0/1, directly connected
S   2001:DB8:3::/64 [1/0]
     via Serial0/0/1, directly connected
C   2001:DB8:4::/64 [0/0]
     via Serial0/0/1, directly connected
L   2001:DB8:4::198/128 [0/0]
     via Serial0/0/1, receive
C   2001:DB8:5::/64 [0/0]
     via GigabitEthernet0/0, directly connected
L   2001:DB8:5::1/128 [0/0]
     via GigabitEthernet0/0, receive
L   FF00::/8 [0/0]
     via Null0, receive
R3#
```

Figure 10: R3 IPv6 Routing Table Output.

4.2 Remediation

All problematic static routes were deleted and replaced with default routes pointing to *R2*'s interface (`172.31.1.197` for IPv4 and `2001:DB8:4::197` for IPv6):

```
! Remove Bad IPv4 Static Routes
R3(config)# no ip route 172.31.0.0 255.255.255.0 Serial0/0/1
R3(config)# no ip route 172.31.1.0 255.255.255.240 Serial0/0/1
R3(config)# no ip route 172.31.1.192 255.255.255.252 Serial0/0/1

! Add IPv4 Default Route
R3(config)# ip route 0.0.0.0 0.0.0.0 172.31.1.197

! Remove Redundant IPv6 Static Routes
R3(config)# no ipv6 route 2001:DB8:2::/64 Serial0/0/1
R3(config)# no ipv6 route 2001:DB8:3::/64 Serial0/0/1

! Add IPv6 Default Route
R3(config)# ipv6 route ::/0 2001:DB8:4::197
```

Listing 3: R3 Corrective Configuration

5 Final Verification

After applying the configuration changes on *R3*, end-to-end ICMP reachability tests were performed across all major network nodes.

Fire	Last Status	Source	Destination	Type	Color	Time(sec)
	Successful	Server	R1	ICMP		0.000
	Successful	PC1	Server	ICMP		0.000
	Successful	PC1	R3	ICMP		0.000
	Successful	PC1	PC2	ICMP		0.000
	Successful	PC2	Server	ICMP		0.000

Figure 11: Final Verification Showing 100% Success Across All Endpoints.

All tests passed successfully, confirming full IPv4 and IPv6 connectivity across the topology.